CSM V3 — BENCH-TEST CHECKLIST

Validate electronics BEFORE installing in machine • Rev V1.0 • ~1-2 h • Page 1: Pre-Test + Stages 1-3

BUILDER: ____________________ **DATE:** ____________________

PRE-TEST SAFETY (REQUIRED)

- ☐ 1. Wall outlet behind a kill switch you can reach without touching the bench.
- ☐ 2. Multimeter ready (DC voltage mode, set to 20 V or 200 V range).
- ☐ 3. No food, drink, or wet items on bench. Dry hands. Insulated mat preferred.
- ☐ 4. E-stop button accessible BEFORE energizing PSU.
- ☐ 5. PSU voltage selector switch matches your wall voltage (115 OR 230 V).

STAGE 1 — PSU ONLY

- ☐ 1. Wire mains: wall → E-stop → 2A fuse → PSU L/N/PE. NOTHING on PSU output.
- ☐ 2. Press E-stop IN. Plug PSU into wall.
- ☐ 3. Release E-stop. PSU green LED should illuminate.
- ☐ 4. Multimeter on PSU +V to -V terminals. Reading: ____ V (expect 24.0 ± 0.2 V)
- ☐ 5. Adjust +V trimpot if needed; lock with paint marker once 24.0 V achieved.
- ☐ 6. Press E-stop. PSU LED off within 1 s. Multimeter to 0 V within 2 s.
- ☐ 7. PASS Stage 1: ☐ YES ☐ NO (If NO: check mains wiring, do not proceed)

STAGE 2 — BUCKS + MEGA

- ☐ 1. E-stop IN. Wire Buck #1 IN+/IN- to 24 V bus. Output disconnected.
- ☐ 2. Release E-stop. Multimeter on Buck #1 output: ____ V (adjust to 5.10 V)
- ☐ 3. E-stop IN. Wire Buck #1 OUT+/OUT- to Mega VIN + GND.
- ☐ 4. Release E-stop. Mega power LED (ON pin) should light.
- ☐ 5. Connect USB to laptop. Mega should enumerate. Upload Blink sketch.
- ☐ 6. LED 13 on Mega blinks at 1 Hz: ☐ verified
- ☐ 7. Multimeter on Mega 5 V pin to GND: ____ V (expect 4.95-5.10)
- ☐ 8. Press E-stop. Mega LEDs off within 1 s.
- ☐ 9. PASS Stage 2: ☐ YES ☐ NO

STAGE 3 — BUCKS #2 + Pi

- ☐ 1. Repeat Stage 2 procedure for Buck #2. Adjust to 5.10 V no-load.
- ☐ 2. Connect Pi 4 via Buck #2 OUT (or USB-C wall wart for bench test).
- ☐ 3. Pi boots, touchscreen shows desktop within 60 s.
- ☐ 4. Multimeter on Pi 5 V pin to GND under load: ____ V (expect ≥ 5.00)
- ☐ 5. PASS Stage 3: ☐ YES ☐ NO

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BUILDER: ______________________ **DATE:** ______________________

STAGE 4 — TB6600 + NEMA 23

- ☐ **1.** E-stop IN. Wire TB6600 power (24 V), step/dir/en to Mega D5/D6/D7 via 220 Ω.
- ☐ **2.** Wire NEMA 23 4-wire to TB6600 A+/A-/B+/B-.
- ☐ **3.** Verify DIP switches: 8 microstep, 2.5 A current limit.
- ☐ **4.** Release E-stop. Upload slow-pulse sketch (1 step/100 ms).
- ☐ **5.** Motor rotates smoothly, no stutter or skipping: ☐ verified
- ☐ **6.** Listen for grinding or whine. Smooth hum is OK; harsh whine = wrong current.
- ☐ **7.** E-stop test: press during rotation. Motor stops within 200 ms.
- ☐ **8.** PASS Stage 4: ☐ YES ☐ NO

STAGE 5 — SERVOS (6× MG90S)

- ☐ **1.** E-stop IN. Wire ONE MG90S to Mega D9 + Buck #1 5 V rail.
- ☐ **2.** Release E-stop. Run sweep sketch (servo position 0-180-0).
- ☐ **3.** Servo motion clean, no jitter: ☐ verified
- ☐ **4.** Repeat for servos on D10, D11, D12, D44, D45. All 6 hold position.
- ☐ **5.** Combined load test: command all 6 servos to move. Multimeter on Buck #1 output.
- ☐ **6.** Voltage stays ≥ 4.95 V during combined motion: ____ V minimum observed
- ☐ **7.** PASS Stage 5: ☐ YES ☐ NO

FINAL VERDICT — Bench Test Complete

☐ ALL 5 STAGES PASS → Electronics validated. Proceed to machine integration.
☐ ANY STAGE FAILED → Fix at the bench, NOT in the machine. Re-test.

NOTES / ANOMALIES: